Technology Data Sheet (TDS)

TINMORRY PC-GF

Version 2.0

• Basic Introduction

TINMORRY PC-GF is a glass fiber reinforced PC composite. It retains the toughness, temperature resistance, and durability of conventional PC, and has significantly improved anti-warping, relatively higher strength, stiffness, layer strength, surface hardness, and wear resistance. It also gives the prints' surface a special texture and feel brought by glass fiber, and hides the layer lines, giving it a delicate and consistent appearance. In addition, it can be printed at very high speed (300 mm/s) on most printers on the market, and has a rich variety of colors to choose from, making it an ideal fiber-reinforced engineering filaments.

Note: 1. Since the PC-GF filament contains glass fiber, please use a nozzle with a diameter of ≥ 0.4 mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them. 2. Please use an enclosed printer to print it to reduce warping and ensure high enough layer strength.

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g ± 0.5%
Length	320 ± 5 m
Spool material	PP ((Temperature resistance: ~ 100 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	85 ± 5 °C，6 - 8 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 75 ℃, ≤ 20% RH （Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set the temp to be about 70 °C ）
Storage temperature and humidity	≤ 30 ℃, ≤ 20% RH（Sealed with desiccant）
Compatible support material	Itself or support filament for PC
Compatible printer type	enclosed-frame (recommended) Open-frame (not recommended)
Compatible nozzle material	Hardened steel
Compatible nozzle size	0.4, 0.6 (recommended), 0.8 mm
Compatible plate type	Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	270 - 300 °C
Bed temperature	90 - 110 °C (depending on the printer and plate types)
Printing speed*	Max: 300 mm/s
Chamber temperature	< 80 °C

*When printed with Bambu X1C and P1S, and the nozzle temperature is 285 °C, the max printing speed can reach 300 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PC-GF have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.30 g/cm^3
Saturated water absorption rate	25 °C , 55% RH	0.23%
Melt index	280 °C, 2.16 kg	18.2 ± 2.1 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	116 °C
Crystallization temperature	DSC, 10 °C/min	N / A
Melting temperature	DSC, 10 °C/min	224 °C
Vicar softening temperature	ISO 306, GB/T 1633	111 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	98 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	107 °C

• Mechanical Properties

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	61 ± 2 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	46 ± 3 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	2752 ± 168 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2046 ± 158 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	3.2% ± 0.8%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	1.4% ± 0.7%
Bending strength (XY)	ISO 178, GB/T 9341	101 ± 3 MPa
Bending strength (Z)	ISO 178, GB/T 9341	73 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	2869 ± 174 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	2118 ± 153 MPa
Impact strength (XY)	ISO 179, GB/T 1043	38.4 ± 2.1 kJ/m²
Impact strength (Z)	ISO 179, GB/T 1043	17.3 ± 2.2 kJ/m²

• Chemical and Other Properties

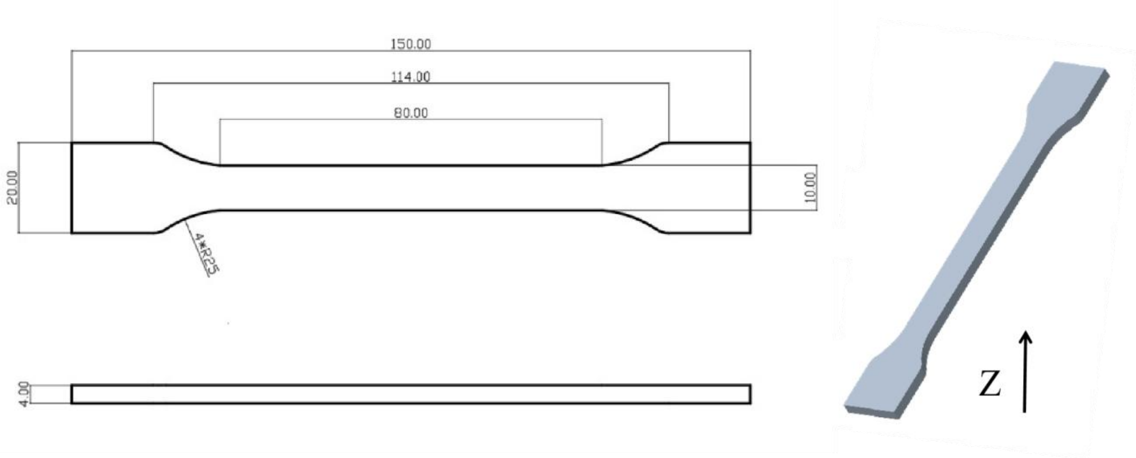
Subjects	Data
Component	PC, glass fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Yes
Resistance to strong acid	Not resistant
Resistance to weak alkali	Yes
Resistance to strong alkali	Not resistant
Flammability	Yes
Odor during printing	Odorless
Odor during combustion	Pungent

• Specimen Info

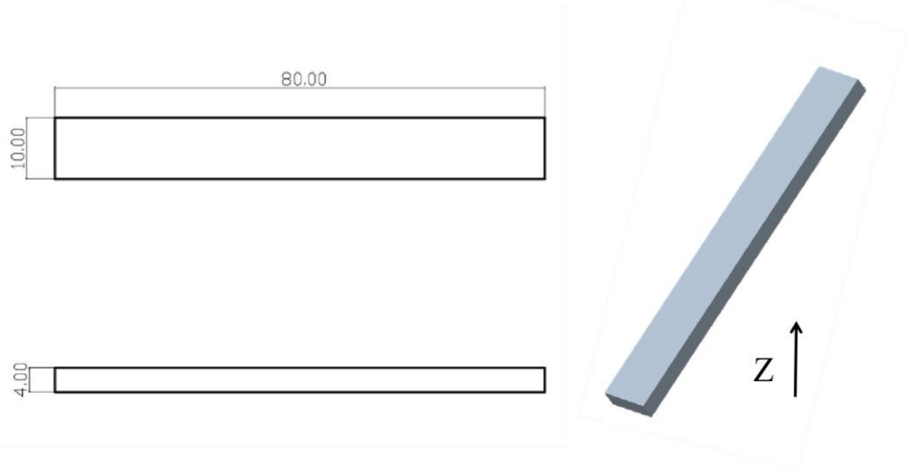
Subjects	Data
Size	Shown at the following pictures
Nozzle temperature	285 °C
Bed temperature	100 °C
Printing speed	XY: 150 mm/s; Z: 80 mm/s
Infill density	100%
Infill pattern	Concentric

Statements：
1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties of TINMORRY PC-GF prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.

1. Tensile Test



2. Bending Test and Impact Test



• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them,or the usage scenarios of them.